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Gourang Chauhan

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Education

Stony Brook University (SUNY) — **August 2025 – Present**
Master of Science in Data/Decision Analytics — *GPA: 3.6*
Courses: Probability and Statistics, Artificial Intelligence and Machine Learning
Graduate Research Assistant: Decision Support Systems

Guru Gobind Singh Indraprastha University — **Aug 2018 – May 2021**
Bachelor's in Business Statistics — *GPA: 8.1*

Skills

Programming Languages	Python, R, Java, C, C++, MATLAB, SQL, Object-Oriented Programming (OOP)
Libraries	TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, MATLAB, Streamlit, Flask
Technologies	Machine Learning, Deep Learning, Computer Vision, Object Detection, Image Segmentation, Image Processing, Explainable AI, TensorFlow Lite
Analytics	Seaborn, Plotly, ETL Pipelines, Business Intelligence, Hypothesis Testing, A/B Testing
Cloud and DevOps	AWS (S3, Lambda, Translate), Google Cloud (Compute Engine, Dialogflow), Firebase

Technical Experience

HSBC Bank Electronic Data Processing — **Gurugram, HR**
Data Scientist — **October 2024 — Aug 2025**

- Built a **RAG-based NLP chatbot** to enable **semantic search** and automated Q&A across **500+ financial compliance documents (50K+ chunks)**, achieving **~90% retrieval accuracy** and **2–4s response time** using **FAISS** and **LLM integration**.
- Developed and deployed an **LLM-powered SAR narrative generation pipeline** using **AWS (SageMaker, Lambda, API Gateway)**, fine-tuned on **15K+ records** to reduce **manual reporting effort by 60%** and improve **AML workflow efficiency**.
- Analyzed and processed **large-scale datasets (50M+ records)** using **PySpark and Python**; automated data workflows with **OpenPyXL**, reducing **reporting time by 70%** and delivering **actionable insights** to stakeholders.

Microsoft Via Talent Software Services — **Noida, IN**
Data Analyst — **Nov 2022 — Oct 2024**

- Analyzed **large-scale structured datasets (100K–1M+ rows)** using **SQL, Excel, and statistical techniques** to generate actionable insights for **Office 365 service performance, usage analytics, and reliability monitoring**, improving **data accuracy by 10–15%** through robust validation and anomaly detection.
- Designed and maintained **Power BI dashboards** for **Office 365 KPIs** (service health, user engagement, operational metrics); automated analytics workflows and optimized **SQL queries and data pipelines**, reducing **manual reporting effort by 20–25%** and improving **query performance by 15–20%.**
- Supported **Office 365 data warehousing and ETL pipelines**, ensuring high-quality data availability for reporting and analytics; achieved **~99% data consistency** through systematic validation, monitoring, and data quality checks..

Wozo Gaming — **Noida, IN**
Data Analyst Intern — **Sep 2021 — Nov 2022**

- Designed and delivered **client-focused projects** by aligning **business requirements**, **user needs**, and **software specifications**; leveraged **Python-based data analysis**, **EDA**, and **statistical forecasting** to drive **20% revenue growth** through data-driven use case design.
- Built and optimized **end-to-end data pipelines** using **AWS (S3, Glue, Athena, Redshift)** with **dimensional modeling**, improving **data retrieval efficiency by 60%**; enhanced **database performance by 35%** using **SQL stored procedures and triggers**, eliminating bottlenecks and improving data workflows.

Projects

Enhancing DINOv3 for Human Gesture Understanding — *Jan 2026 to Mar 2026*
Self-Supervised Learning, ViT, LoRA

- Built a **multimodal deep learning framework** using **CNNs and ensemble models** for **ARMD diagnosis**, achieving **99.41% (OCT) and 97.70% (fundus) accuracy**; published a **12K+ labeled dataset** and developed a **Grad-CAM pipeline**, improving **model explainability by 28%.**

Spotify Song Popularity Prediction — *Oct 2025 to Dec 2025*
Python, Pandas, NumPy, Scikit-Learn, XGBoost, TensorFlow/Keras

- Built a **machine learning model** to predict song popularity pre-release, preprocessing a **600K-song dataset** with scaling, encoding, and artist popularity classification to help optimize music marketing strategies.
- Achieved **R2 = 79.5%** using **XGBoost, Neural Networks, and Random Forest,** identifying **valence** and **energy** as key predictive factors to improve audience targeting and engagement insights.

FinAssist AI: Financial Product GenAI Chatbot — *July 2024 to Aug 2024*
Python, LangChain, HuggingFace, Zephyr-7B

- Built a **Retrieval-Augmented Generation (RAG) chatbot** for **financial product query answering** using **LangChain** and **Zephyr-7B**; generated **text embeddings** with HuggingFace and enabled **semantic search** using **FAISS**.
- Preprocessed and structured **raw product and FAQ data** into CSV format; tested and optimized the **RAG pipeline** by evaluating response relevance and improving retrieval accuracy.